HALO: Hybrid Attention Model for Subcellular Localization

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Abstract

Subcellular localization prediction is essential for understanding protein function and guiding therapeutic discovery. We introduce **HALO** (Hybrid Attention model for subcellular LOcalization), a hybrid framework that integrates protein language model embeddings from fine-tuned ESM with structural information from AlphaFold and biochemical amino acid properties. HALO employs graph attention networks to unify sequence, structural, and biochemical features, with a learnable weighting scheme to balance contributions across modalities. To address class imbalance, we implement a focal loss with adaptive thresholding. Evaluated on three low-homology datasets, HALO achieves state-of-the-art performance, effectively predicting localization for proteins with or without structural data. This highlights the potential of combining PLMs and structural embeddings in graph-based neural networks, providing a flexible and generalizable framework for protein function annotation.

Introduction

Existing subcellular localization tools have clear gaps: alignment-based methods depend on close homologs and propagate annotation bias; feature-engineered ML (e.g., amino-acid composition/ physicochemical profiles) misses long range, context dependent interactions; sequence-only deep models overlook 3D geometry, membrane topology, disorder, and targeting signals; single modality systems degrade when structure or high-quality sequence cues are unavailable; benchmarks often suffer from homology leakage, severe class imbalance, weak multi-label handling, and poor calibration/cross-species transfer. **HALO** addresses these in one shot by fusing fine-tuned PLM embeddings with AlphaFold-derived structural features and biochemical properties via a GAT, using learnable weights to stay robust when a modality is missing, employing focal loss with adaptive thresholding for imbalance/multi-label calibration, and training on low-homology splits to ensure realistic generalization.

Importing Tables & Graphs

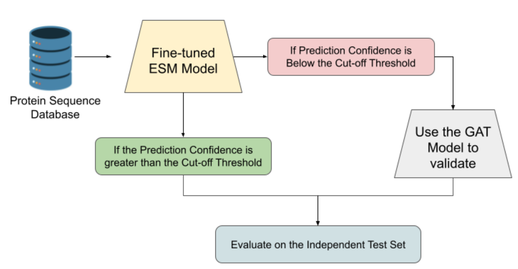

Figure : A hybrid model leveraging Fine-tuned ESM and AlphaFold structure-based GAT attention model for subcellular localization.

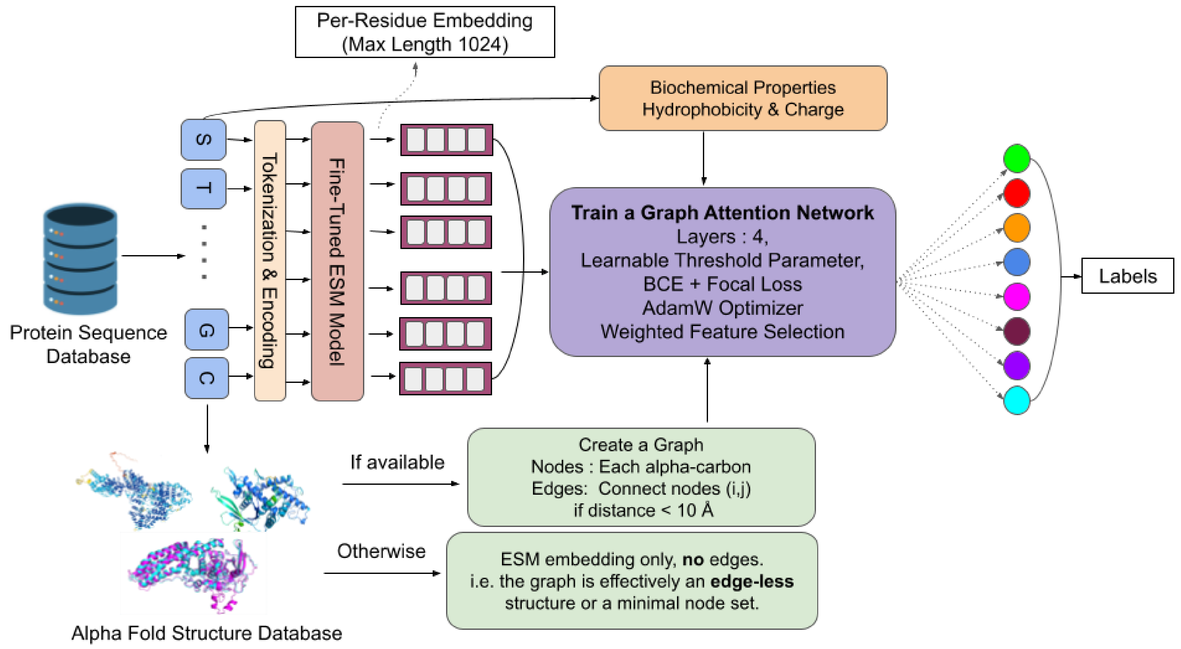

Figure : Leveraging the Fine-Tuned ESM with AlphaFold Structures and Biochemical Properties for training a weighted graph attention model.

Methodology

The architecture of HALO combines 2 parts –

1. Fine-tuning ESM for the localization task.
2. Applying graph attention network (GAT)
3. Combining fine-tuned ESM with GAT using a hybrid method using the confidence analysis. (HALO).

Fine-Tuned ESM

We fine-tune an ESM2 backbone for multi-label localization: sequences (≤**1024 aa**) are tokenized, passed through ESM2, and summarized by an attention aggregator that learns token weights to form a single sequence embedding; a linear head outputs per-class logits. Most ESM2 layers are frozen (last two encoder blocks trainable) to control compute and overfitting. Training uses focal loss to emphasize hard examples and updates only the unfrozen ESM2 blocks, attention head, and classifier. We log checkpoints, monitor macro-F1, and perform threshold tuning (0.10–0.95) post-training to pick a deployment threshold (best at 0.45 on validation).

Graph Attention

For structure-aware learning, we extract **per-residue ESM embeddings** and pair them with **AlphaFold** 3D coordinates when available; sequences without structures are still used (node features only). Each protein becomes a graph: **Cα nodes**, edges if distance < **10 Å**. Node features concatenate **scaled coordinates**, **fine-tuned ESM residues**, and **biochemical properties**. A multi-layer **GAT** learns attention over residue–residue edges; **global mean pooling** yields a graph embedding for multi-label classification via focal loss with a **learnable threshold**. We sweep **feature weights** (coord/esm/bio) to balance modalities and select the combo maximizing validation macro-F1 under compute limits.

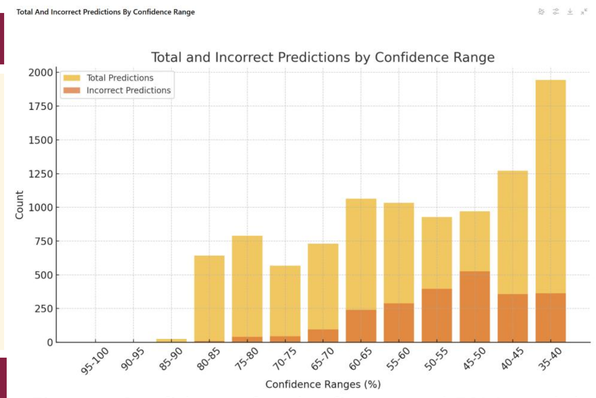

Figure : Confidence for the fine-tuned ESM model.

Results

Metric	ESM	F-ESM	WF-ESM
Macro F1	0.5420	0.55	0.5942
Micro F1	0.6023	0.61	0.6803
Accuracy	0.2671	0.24	0.3819
Jaccard Index	0.5375	0.5375	0.6164
Precision	0.4706	0.48	0.5771
Recall	0.8361	0.85	0.8283
ROC AUC	0.9293	0.93	0.9399

Table : Performance comparison of three GAT models on the DeepLoc test set.

Metric	HALO	Fine-tuned ESM	GAT
Macro F1	**0.6782**	0.6688	0.5942
Micro F1	0.7364	**0.7471**	0.6803
Accuracy	**0.5756**	0.5621	0.3819
Jaccard Index	**0.7187**	0.7075	0.6164
Precision	**0.7098**	0.6996	0.5771
Recall	0.6511	0.6537	**0.8283**
ROC AUC	**0.9412**	0.9254	0.9399

Matthews Correlation Coefficient (MCC) per Label

Label	HALO	Fine-Tuned ESM	GAT
Cytoplasm	0.6125	**0.6162**	0.4798
Nucleus	0.6819	**0.6821**	0.5731
Extracellular	**0.8559**	0.8554	0.8389
Cell Membrane	**0.6752**	0.6660	0.5401
Mitochondrion	0.7921	**0.7999**	0.7371
Plastid	**0.8918**	0.8758	0.8456
Endoplasmic Reticulum	0.5735	**0.5864**	0.5314
Lysosome/Vacuole	0.2621	**0.2982**	0.2674
Golgi Apparatus	**0.3349**	0.3283	0.3212
Peroxisome	0.5835	**0.5991**	0.3112

Table : Comparison of overall performance metrics and MCC per subcellular localization label across HALO, fine-tuned ESM, and GAT models on the DeepLoc test set.

MCC	HALO_S	RF_S	MLP_S	HALO_T	RF_T	MLP_T	HALO_N	RF_N	MLP_N
Cell Junction	0	0.37	0.187	0	0.06	0.07	0	0	0
Cell Membrane	0.83	0.78	0.824	0.72	0.62	0.67	0.5	0.21	0.26
Cell Projection	0	0.069	0.151	0	0.21	0.12	0	0	0
Cytoplasm	0.66	0.558	0.620	0.44	0.40	0.43	0.26	0.19	0.19
ER	0.71	0.618	0.699	0.74	0.65	0.71	0.58	0.19	0.18
Golgi Apparatus	0.62	0.605	0.612	0.43	0.51	0.48	0.34	0.41	0.25
Lysosome	0.24	0.380	0.314	0.14	0.68	0.63	0	0	0
Mitochondrion	0.89	0.878	0.865	0.77	0.73	0.64	0.67	0.47	0.23
Nucleus	0.87	0.730	0.836	0.76	0.61	0.68	0.55	0.26	0.30
Secreted	0.91	0.866	0.896	0.81	0.69	0.72	0.60	0.53	0.52

Table : Comparison across the three datasets S (Swiss-Prot), T (TrEMBL Independent), and N (non-homology TrEMBL Independent) for our HALO model and the interpretable LLM models.

HALO

HALO routes every sample through ESM first; labels with **confidence ≥55%** are accepted. For labels **<55%**, we consult the **GAT**: if ESM and GAT **agree**, the label is kept; if not, it's discarded, yielding conservative, precision-oriented multi-label outputs. This cross-checking fuses **sequence semantics** (ESM) with **structural adjacency** (GAT), improving precision across low-confidence regimes while preserving ESM's high-confidence strengths resulting in **more calibrated** and **robust** localization calls than either model alone.

Conclusion

HALO is a hybrid method that fuses fine-tuned LLM sequence embeddings, biochemical features (e.g., charge, hydrophobicity), and AlphaFold-derived structure to improve subcellular localization across datasets showcasing the value of multi-modal protein annotation.

Acknowledgements

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